

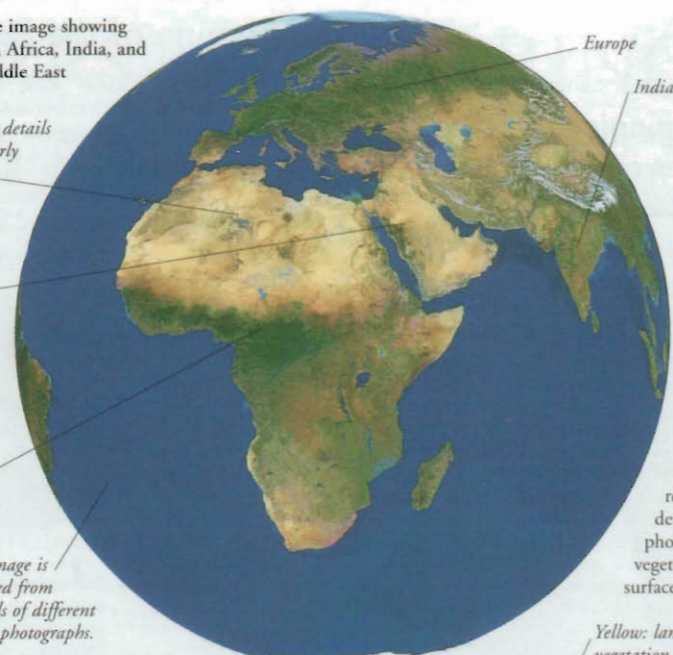
Satellite image showing Europe, Africa, India, and the Middle East

Surface details are clearly visible.

Middle East

Africa

Large image is assembled from hundreds of different satellite photographs.



Biosphere

Between the atmosphere's lowest layers and the ocean floor is a rich diversity of life, from tiny ocean organisms called plankton to the largest trees and animals. Together, these organisms form the biosphere – the living part of the planet. Satellite images can help scientists understand the complicated links between living things and the Earth.

Green: land areas where vegetation is most dense

Infrared image of vegetation and plankton distribution



Yellow: land areas where vegetation is least dense

Red: ocean areas where plankton is most dense

Locator map

Blue: ocean areas where plankton is least dense

Earth system

Planet Earth seems to operate like a vast, complex system made up of various interconnected processes that keep conditions stable and suitable for life. The atmosphere's unique make up, for example, ensures that the Earth stays at an ideal temperature for life, never heating up or cooling down by more than a few degrees. Scientists now realize that the environment must be treated with care, because a change to one part of this complex system may have unpredictable repercussions in other parts.

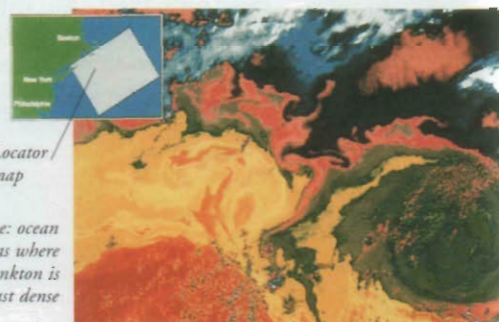
Earth from space

Much of what scientists know about the interrelated parts of the Earth system comes from images sent back by satellites. These images show us that the Earth is roughly spherical and reveal surface features in clear detail. Special heat-sensitive infrared photographs show the distribution of vegetation and variations in the Earth's surface temperature.

Energy regulation

The Earth system exchanges energy with its surroundings, but there is no overall gain or loss of energy. The Earth receives heat, light, and other forms of energy directly from the Sun. Some of this energy is reflected back by the clouds, oceans, land, and atmosphere; the rest is absorbed and then released back into space. The total energy the Earth gives out equals the total energy it receives from the Sun.

Infrared image of temperature variations in the Atlantic Ocean off the USA's eastern coast



Gaia theory

British scientist James Lovelock (b. 1919) suggests that the Earth and all the lifeforms upon it function as if they were a single living organism. He calls this "organism" Gaia, after the Greek goddess of fertility. Like any other organism, he says, Gaia is self-regulating, meaning that it will naturally change its environment to maintain the right conditions for life – even if humans make the Earth unfit for themselves by polluting it and using up its limited resources.



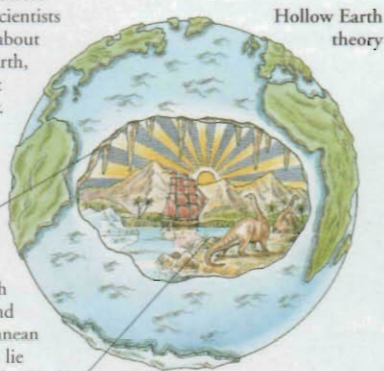
Greek statue of Gaia, 450 BC

Theories about the Earth

There have been many theories about the Earth that may seem strange to people today, but which were widely believed at the time. The ancient Egyptians, for example, thought that the Earth was a flat square under a pyramid-shaped sky, and people in medieval Europe believed that it was the Sun that revolved around the Earth, and not vice versa. Similarly, before technology enabled scientists to understand more about the interior of the Earth, people suggested that the Earth was hollow.

People assumed the Earth had a vast, empty core.

Hidden lands and oceans, complete with plants and animals and warmed by a subterranean Sun, were thought to lie within the centre of the Earth.



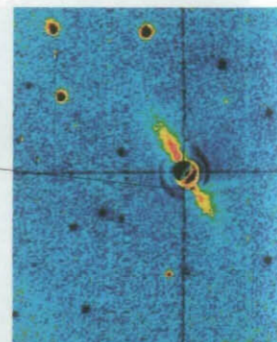
Hollow Earth theory

Search for another Earth

Astronomers have recently detected signs of the existence of planets beyond the Solar System. Wobbles in the movements of the stars 47 Ursae Majoris, 70 Virginis, and 51 Pegasi suggest that they may be orbited by planets – perhaps even ones similar to the Earth. Astronomers have found other stars with solar systems forming around them.

The yellow-and-red area may be another solar system forming around Beta Pictoris.

False-colour satellite image of the star Beta Pictoris, about 50 light years away



Timeline

c.4,600 mya The Earth and the other planets form as parts of a vast cloud of hot gas and dust circling the Sun begin to cluster together.

c.4,300 mya The Earth's crust forms.

c.4,200 mya As the Earth cools, gas bubbles and water vapour rise from the interior to form a cloudy atmosphere.



Gneiss rock

c.4,000 mya The crust and mantle separate; rain begins to fall; the atmosphere clears.

c.3,800 mya The first organisms are single-celled bacteria.

c.3,000 mya The atmosphere becomes oxygen-rich as ocean plants absorb sunlight and release oxygen into the air.

c.1,500 mya Protists, such as amoeba, are the first complex living cells; later, protists join up to form sponges – the first multi-celled organisms.

c.570 mya A huge variety of complex lifeforms develops in the Earth's seas and oceans.



Sponge

c.440–400 mya Land-based plants and animals become widespread.

c.220 mya There is a single, vast land mass, now known as Pangaea, which later breaks up into the smaller land masses we today call continents.

c.200–70 mya The era of the dinosaurs.

c.100,000 ya First modern humans appear.

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